worked well last winter when temperatures fell to freezing. By combining all of these facts about the physical system together in my mind, I decide that it's time to book in a service of my air conditioner unit—I have integrated multiple sources of knowledge together to determine what action I should take.

The challenge is that the additional facts that I have used to determine this action (the unit is set to 24°, the service manual recommends refrigerant gas, my last service was two years ago, the unit has previously handled cool temperatures) are currently all in my mind, and are not represented in a systematic way that my digital twin can reason about.

5.1.1 From measurements to decisions

Digital twins must transform raw sensor readings into actionable decisions. This progression is captured by Ackoff's pyramid, a framework developed by organizational theorist Russell Ackoff that shows how data gains value through successive layers of refinement. As shown in Figure 5.2, the pyramid has four levels: data becomes information when given context, information becomes knowledge when patterns and relationships are understood, and knowledge becomes wisdom when applied to make sound decisions.

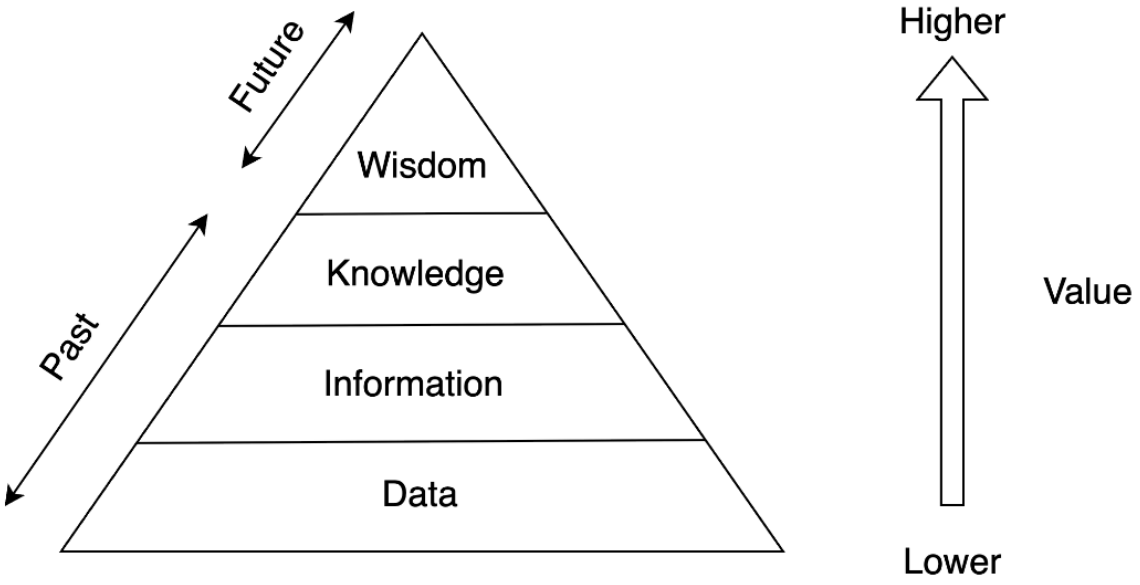


Figure 5.2 Ackoff's pyramid showing how valuable knowledge and wisdom are less available than lower value data and information.